WORKSHEET

Chapter 5: Parametric and Cartesian equations

1 Find the Cartesian equation that corresponds to each of the following vector equations.

a $\mathbf{r}(t) = (t - 1)\mathbf{i} + (t + 1)\mathbf{j}\ (t \in R)$

b $\mathbf{r}(t) = (2t + 1)\mathbf{i} + (t - 2)\mathbf{j}\ (t \in R)$

c $\mathbf{r}(t) = (1 - t)\mathbf{i} + (3 - 5t)\mathbf{j}\ (t \in R)$

d $\mathbf{r}(t) = (1 - t^2)\mathbf{i} + (2t^2 - t^4)\mathbf{j}\ (t \in R)$

e $\mathbf{r}(t) = 2t\mathbf{i} + (t^2 - 1)\mathbf{j}\ (t \in R)$

f $\mathbf{r}(t) = (2t - 1)\mathbf{i} + (t^2 + t)\mathbf{j}\ (t \in R)$

2 State the domain and range of the each Cartesian relation found in Question 1.

a Domain:

Range:

b Domain:

Range:

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c Domain:

Range:

d Domain:

Range:

e Domain:

Range:

f Domain:

Range:

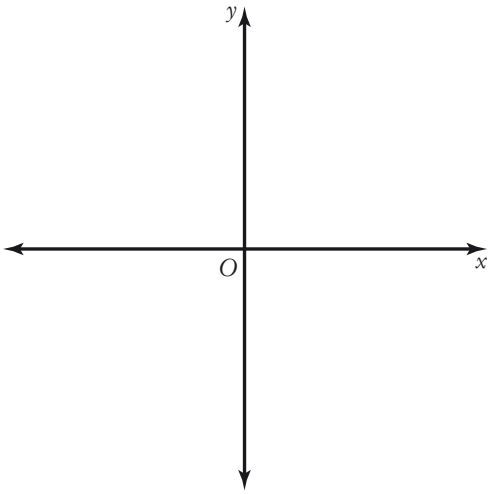
3 A number of position vector functions is given below. For each function:

i find the corresponding Cartesian equation

ii draw a sketch of the path of the particle indicating the direction of travel

iii state the domain and range of the Cartesian function.

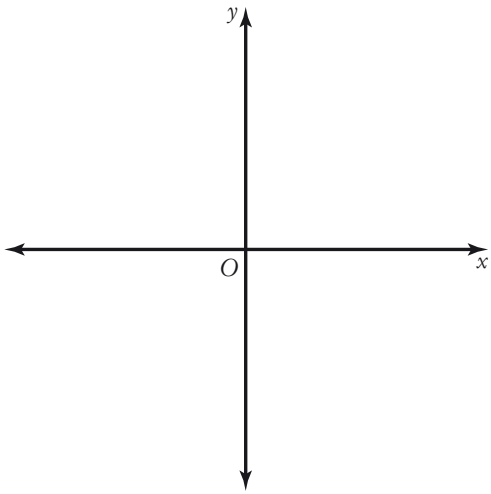
a $\mathbf{r}(t) = (2 - t)\mathbf{i} + (t + 6)\mathbf{j}$ for $0 \leq t \leq 6$



Domain:

Range:

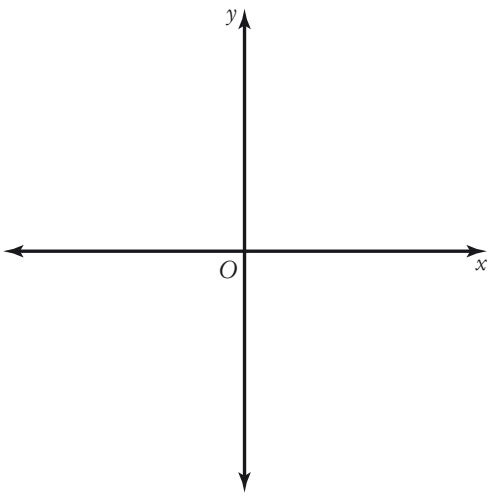
b $\mathbf{r}(t) = (2 - t)\mathbf{i} + (3 + t^2)\mathbf{j}$ for $-1 \leq t \leq 3$



Domain:

Range:

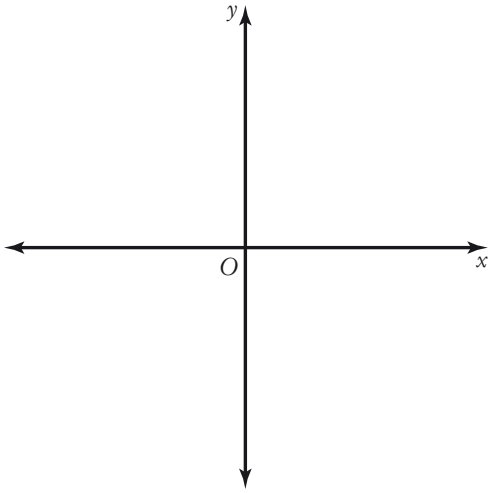
c $\mathbf{r}(t) = 2t\mathbf{i} + (t^2 - 4)\mathbf{j}$ for $-4 \leq t \leq 3$



Domain:

Range:

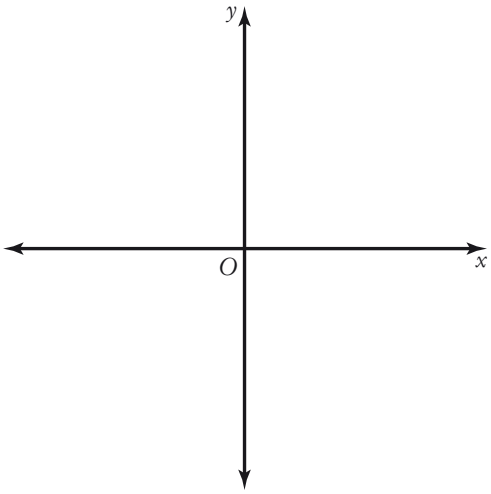
d $\mathbf{r}(t) = (8t^2 - 5)\mathbf{i} + (4t - 1)\mathbf{j}$ for $-1 \leq t$



Domain:

Range:

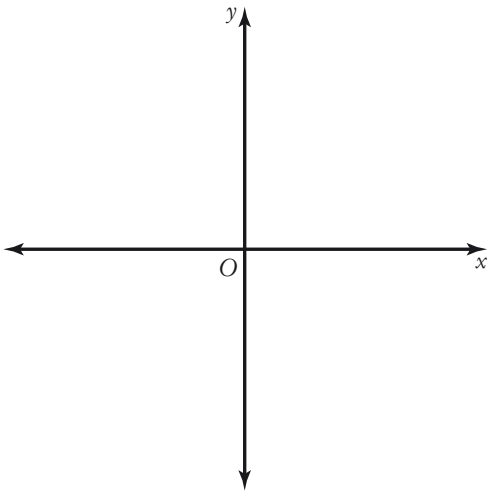
e $\mathbf{r}(t) = (2t - 3)\mathbf{i} + (4t^2 - 7)\mathbf{j}$ for $t \in R$



Domain:

Range:

f $\mathbf{r}(t) = \frac{1}{t}\mathbf{i} + \frac{1}{t+1}\mathbf{j}$ for $t > 0$



Domain:

Range:

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4 Each of the following represents a position vector function for a particle. Find the Cartesian relation that corresponds to the function in each case.

a $\mathbf{r}(t) = 5\cos(t)\mathbf{i} + 2\sin(t)\mathbf{j}$ for $t \in [0, 2\pi]$

b $\mathbf{r}(t) = \cos(2t)\mathbf{i} + \cos(t)\mathbf{j}$ for $t \in [0, 2\pi]$

c $\mathbf{r}(t) = \sin^2(t)\mathbf{i} + 2\cos(t)\mathbf{j}$ for $t \in [0, 2\pi]$

d $\mathbf{r}(t) = 2\cos(t)\mathbf{i} + 3\sin(t)\mathbf{j}$ for $t \in \left[0, \frac{\pi}{2}\right]$

e $\mathbf{r}(t) = \sin(t)\mathbf{i} + \cos(2t)\mathbf{j}$ for $t \in [0, 2\pi]$

f $\mathbf{r}(t) = \sin^2(t)\mathbf{i} + \cos(2t)\mathbf{j}$ for $t \in \left[0, \frac{\pi}{2}\right]$

5 State the domain and range of each of the relations found in Question 4.

a Domain:

Range:

b Domain:

Range:

c Domain:

Range:

d Domain:

Range:

e Domain:

Range:

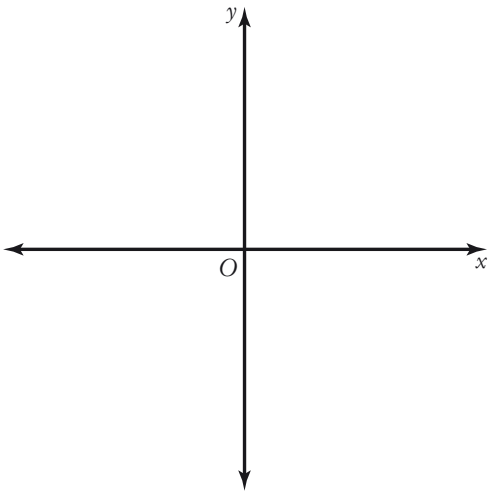
f Domain:

Range:

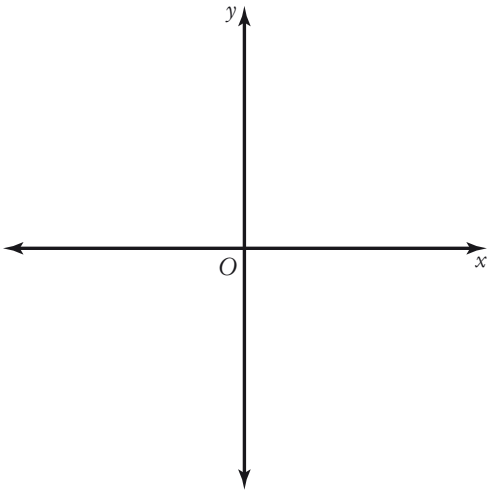
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6 For each of the relations found in Question 4, draw a graph of the path of the particle indicating the direction of travel.

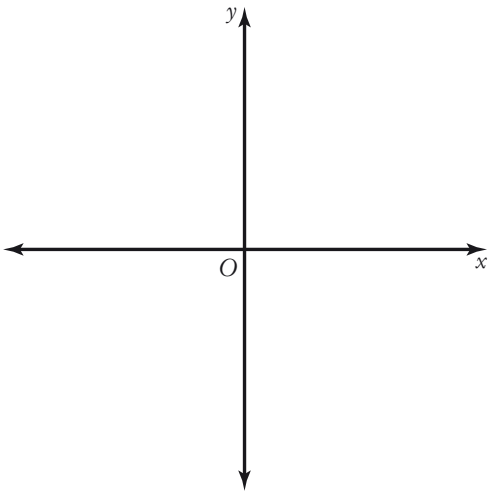
a



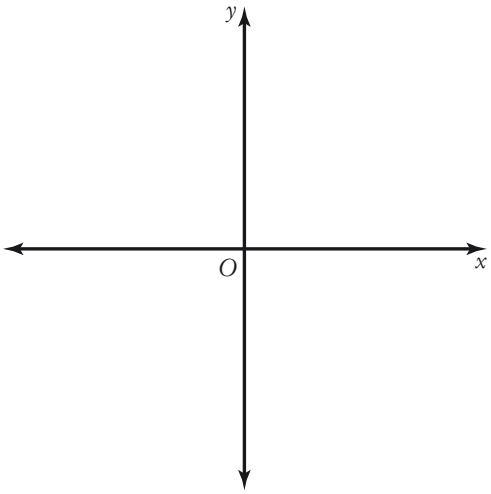
b



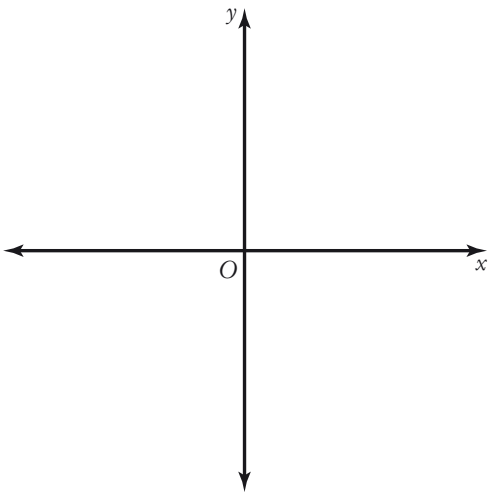
c



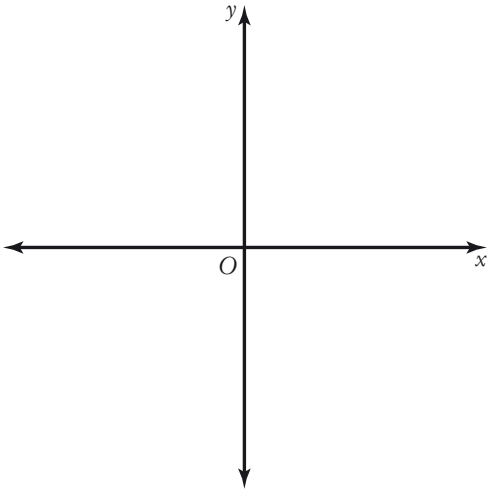
d



e



f



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7 The positions of two particles are defined by the vector functions:

$\mathbf{r}(t) = (2t - 3)\mathbf{i} + (t^2 + 8)\mathbf{j}$

$\mathbf{q}(t) = (t + 1)\mathbf{i} + 6t\mathbf{j}$

a Find the point at which the particles collide.

b Find the points where the paths of the particles cross.

c Calculate the distance between the particles when $t = 1$.

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Answers

1 a $y = x + 2$

b $y = \frac{x}{2} + \frac{5}{2}$

c $y = 5x - 2$

d $y = 1 - x^2$

e $y = \frac{x^2}{4} - 1$

f $y = \frac{x^2}{4} + x + \frac{3}{4}$

2 a Domain: $x \in \boldsymbol{R}$, Range: $y \in \boldsymbol{R}$

b Domain: $x \in \boldsymbol{R}$, Range: $y \in \boldsymbol{R}$

c Domain: $x \in \boldsymbol{R}$, Range: $y \leq 1$

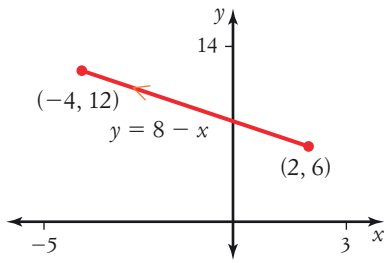
d Domain: $x \in \boldsymbol{R}$, Range: $y \leq 1$

e Domain: $x \in \boldsymbol{R}$, Range: $y \geq 1$

f Domain: $x \in \boldsymbol{R}^+$, Range: $y \geq \frac{-1}{4}$

3 a i $y = 8 - x$

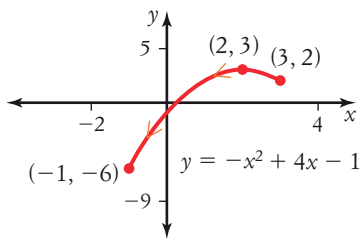
ii



iii Domain: $-4 \leq x \leq 2$; Range: $6 \leq y \leq 12$

b i $y = -x^2 + 4x - 1$

ii

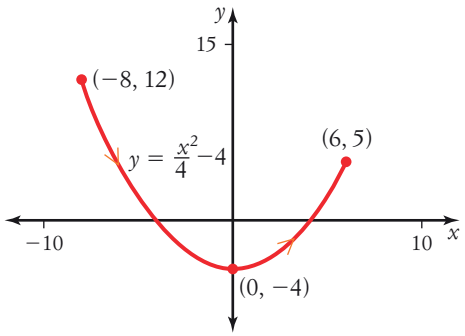


iii Domain: $-1 \leq x \leq 3$; Range: $-6 \leq y \leq 3$

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c **i** $y = \frac{1}{4}x^2 - 4$

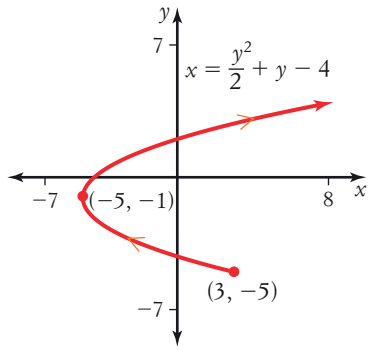
ii



iii Domain: $-8 \leq x \leq 6$; Range: $-4 \leq y \leq 12$

d **i** $x = \frac{1}{2}y^2 + y - 4$

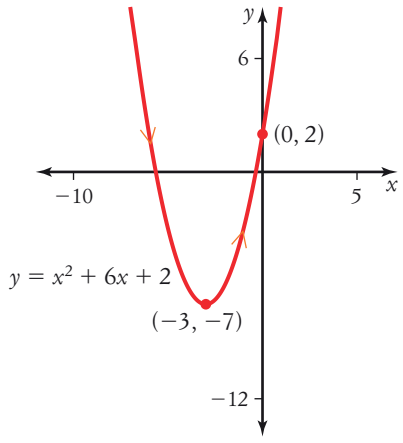
ii



iii Domain: $-5 \leq x$; Range: $-5 \leq y$

e **i** $y = x^2 + 6x + 2$

ii

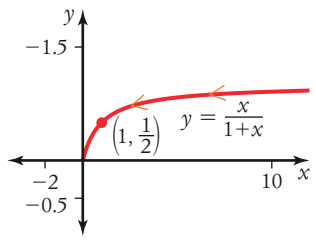


iii Domain: $x \in \boldsymbol{R}$; Range: $-7 \leq y$

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f **i** $y = \dfrac{x}{1+x}, x \neq 0, 1$

ii



iii Domain: $x > 0$; Range: $y > 0$

4 **a** $\dfrac{x^2}{25} + \dfrac{y^2}{4} = 1$

b $x = 2y^2 - 1$

c $x = 1 - \dfrac{y^2}{4}$

d $\dfrac{x^2}{4} + \dfrac{y^2}{9} = 1$

e $y = -2x^2 + 1$

f $y = 1 - 2x$

5 **a** Domain: $-5 \leq x \leq 5$; Range: $-2 \leq y \leq 2$

b Domain: $-1 \leq x \leq 1$; Range: $-1 \leq y \leq 1$

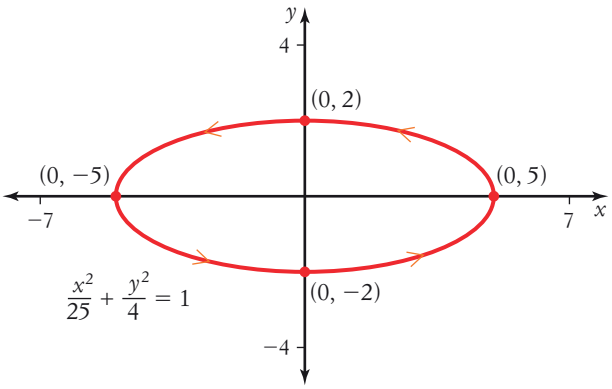
c Domain: $0 \leq x \leq 1$; Range: $-2 \leq y \leq 2$

d Domain: $0 \leq x \leq 2$; Range: $0 \leq y \leq 3$

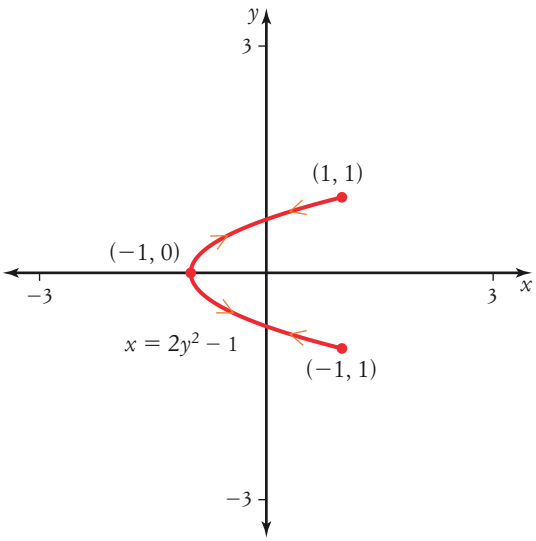
e Domain: $-1 \leq x \leq 1$; Range: $-1 \leq y \leq 1$

f Domain: $0 \leq x \leq 1$; Range: $-1 \leq y \leq 1$

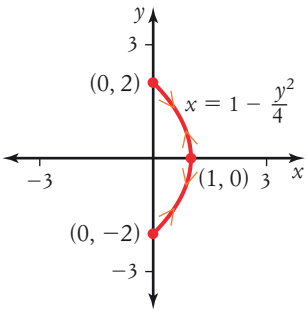
6 **a**



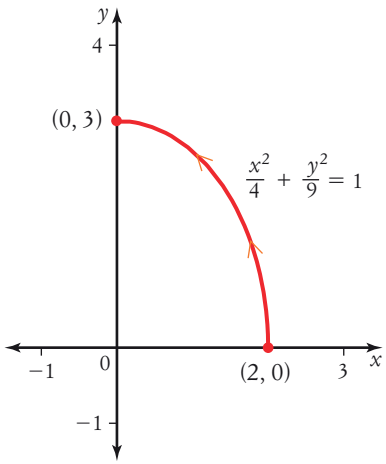
b



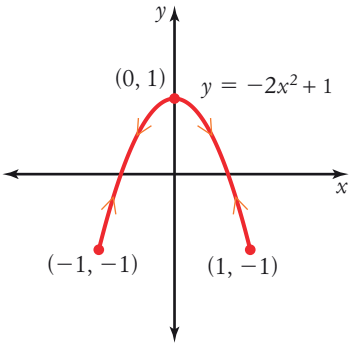
c



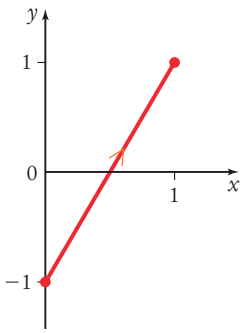
d



e



f



7 **a** (5, 24)

b (5, 24) and (13, 72)

c $3\sqrt{2}$

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